

Verdant v1 — Battery-Powered Plant Waterer

Purpose

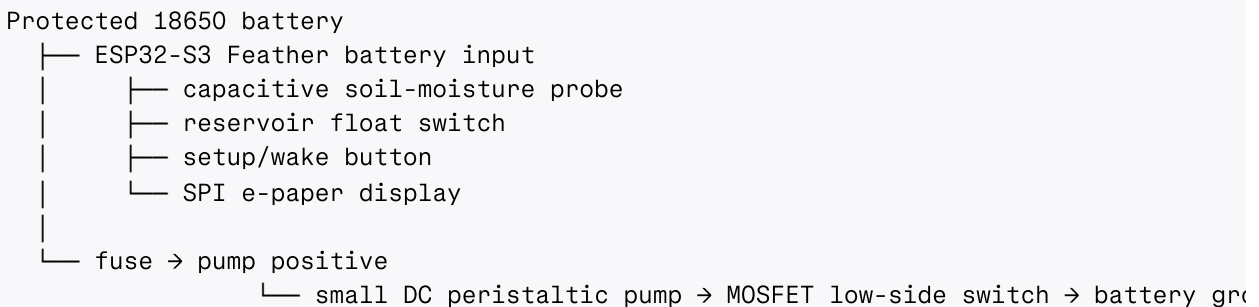
Verdant v1 is a single-potted-plant automatic watering controller with:

- ESP32-S3 firmware written in C++ using VS Code and PlatformIO
- A removable protected 18650 battery
- Soil-moisture-based watering decisions
- A small peristaltic pump controlled through a MOSFET
- Reservoir-empty protection with a float switch
- A small e-paper screen for moisture, battery, tank, and pump status
- A local web UI available only during a short, intentional Wi-Fi setup session
- A dry, gasketed enclosure physically separated from water tubing and the reservoir

This guide is for **one plant / one pump / one battery**. It deliberately avoids mains voltage, relays, 12 V pumps, and always-on Wi-Fi.

Before You Buy

Architecture



The controller sleeps most of the time. Every 30–60 minutes it wakes, reads the sensor, checks the reservoir and battery state, optionally runs a small timed pump dose, updates the e-paper display only when useful, and returns to deep sleep.

Buy List

1. Controller and battery

Qty	Item	Required specification	Notes
1	Adafruit ESP32-S3 Feather	Standard non-display board; USB-C; Wi-Fi/BLE; single-cell Li-ion/LiPo charging; JST-PH battery connector	Recommended controller for Verdant v1
2	Protected 18650 Li-ion cells	Genuine 3.6/3.7 V nominal, 4.2 V maximum, 2,500–3,500 mAh; ideally same model and purchase batch	One installed, one charged spare for battery swaps
1	Single-cell 18650 holder with leads	Must fit a protected/button-top cell, which is longer than an unprotected 18650	Do not assume every holder fits protected cells
1	JST-PH 2-pin pigtail	Correct connector for Feather battery input	Confirm polarity with a multimeter before connecting
1	External 18650 charger	Reputable 1- or 2-bay Li-ion charger	Use for routine charging indoors
1	USB-C data cable	Must support data, not charge-only	Programming and optional in-device charging
1	SPST DC-rated power switch	At least 1 A DC	Physical battery disconnect

2. Pump and switching electronics

Qty	Item	Required specification	Notes
1	Micro peristaltic pump	Rated to run from approximately 3–4.2 V; low-flow; self-priming	Avoid a 12 V pump and avoid a 5–6 V-only pump for this one-cell design
1	N-channel MOSFET motor driver module	Explicitly accepts 3.3 V logic; low-side DC motor switching; at least 1 A continuous and 2 A peak	Use this instead of a relay
1	Flyback diode	1N5819 preferred or 1N4007	Not needed if the MOSFET module explicitly includes a diode
1	Inline fuse holder	Suitable for the battery-to-pump positive wire	Put the fuse as close to battery positive as practical
2–3	Fuses	Start with 1 A; finalize after measuring pump stall current	Never select only from normal running current
1	Electrolytic capacitor	470–1,000 μ F; at least 6.3 V, preferably 10 V	Across pump supply near MOSFET/pump to reduce voltage dips
1	Ceramic capacitor	0.1 μ F	Across pump terminals to reduce motor noise

3. Sensors and local controls

Qty	Item	Required specification	Notes
1	Capacitive analog soil-moisture sensor	3.3 V-compatible analog output	Do not buy exposed-metal resistive two-prong probes
1	Float switch	Vertical or horizontal based on reservoir geometry	Hardware reservoir-low interlock; wire to a GPIO using INPUT_PULLUP

Qty	Item	Required specification	Notes
1	Momentary push button	Normally open	Physical setup/wake button for a temporary Wi-Fi/web UI session
Optional	Leak sensor	Two-contact water-leak sensor or conductive trace sensor	Mount below/near the electronics enclosure; use it to lock out pump operation
Optional	MAX17048 fuel-gauge breakout	I ² C single-cell Li-ion fuel gauge	Only needed if the chosen Feather's integrated battery reporting is not sufficient

4. E-paper display

Qty	Item	Required specification	Notes
1	Monochrome e-paper/e-ink SPI module	1.54-inch, black/white, 200 × 200, 3.3 V-compatible SPI interface; partial refresh preferred	Recommended status display
1	Clear polycarbonate or acrylic window	Sized for display opening	Goes in front of display in enclosure
1	Thin silicone/EPDM gasket sheet	Enough to make a window gasket	Do not rely only on hot glue
1	Printed display bezel or small frame	3D print	Compresses the window gasket against the enclosure

A suggested screen layout:

```

VERDANT
Battery: 82% 3.91 V
Soil: 36%
Tank: OK
Pump: OFF
Last: 6h ago

```

Update the e-paper display only on a watering event, a fault/state change, meaningful battery/moisture change, or an occasional heartbeat. Call the display sleep/hibernate method after updating.

5. Water system

Qty	Item	Required specification	Notes
1	Opaque lidded water reservoir	Approximately 0.5–2 L	Opaque material reduces algae; lid reduces contamination and spills
1	Silicone tubing	Match the pump barb's exact ID/OD	Buy extra for routing and replacements
1	Inline check valve	Match tubing size	Strongly recommended to prevent siphoning/backflow
1–2	Drip stake, tube holder, or clip	Fits tubing	Keep outlet fixed and above the soil surface

Qty	Item	Required specification	Notes
1	Intake filter / fine mesh	Fits intake arrangement	Optional but useful to keep debris out of tubing/pump
1	Catch tray / secondary containment	Large enough for pot and/or reservoir	Required during testing; recommended permanently

6. Water-resistant enclosure and mounting

Qty	Item	Required specification	Notes
1	Commercial gasketed electrical enclosure	Polycarbonate or ABS, IP65 minimum; IP67 preferred if temporary immersion is plausible	This is the primary electronics water barrier—not a bare FDM print
2–4	Cable glands	IP67-rated nylon PG7/PG9 or metric equivalent; sized to actual cable diameter	Put entries on bottom-facing side of box
1	Neutral-cure RTV silicone or polyurethane sealant	Electronics-safe / neutral cure	Seal non-serviceable window edge and fixed penetrations
1–2	Desiccant packets	Small replaceable packets	Helps reduce condensation inside enclosure
1	Printed mounting bracket	3D print	Mount box high above reservoir and catch tray
1	Printed pump bracket	3D print	Keep pump stable and separate from dry electronics box
1	Printed float-switch bracket	3D print as needed	Keeps switch positioned consistently
1	Battery retainer	3D print	Prevents 18650 from moving in holder
Optional	Breather vent	Rated waterproof membrane vent	Useful for outdoor temperature swings; do not add an open vent hole

Mounting rules

- Keep the electronics box above the highest possible reservoir water level.
- Keep all water tubing outside the dry electronics enclosure.
- Put cable glands on the bottom or downward-facing side.
- Add a drip loop to every cable before it enters a gland.
- Put electronics on internal standoffs rather than on the enclosure floor.
- Keep pump wiring separate from the moisture sensor’s analog cable when possible.
- Twist the pump’s positive/negative pair to reduce electrical noise.

7. Wiring and assembly materials

Qty	Item	Notes
1	Mini breadboard	Temporary dry prototyping only—not permanent battery/pump wiring
1	Perfboard or prototyping PCB	Permanent soldered assembly
1 spool	22–24 AWG stranded wire	Battery and pump wiring
1 spool	26–28 AWG stranded wire	Sensor/display/button signal wiring
1 set	Dupont jumpers	Prototyping convenience
1 kit	Heat-shrink tubing	Insulate all battery and pump splices
1 kit	Crimp terminals/connectors	Make pump, sensor, and display wiring serviceable
1 set	Cable ties and adhesive tie mounts	Strain relief and wire management
1 pack	M2/M2.5/M3 screws, nuts, and standoffs	Secure modules and printed brackets
1 roll	Double-sided foam mounting tape	Optional for noncritical cable management
1 set	Labels	Label battery polarity, fuse rating, and service instructions

8. Tools

Tool	Why it is needed
Multimeter	Required: check polarity, voltage, continuity, and pump current
Soldering iron and solder	Durable permanent wiring
Wire stripper/cutter	Prepare wire safely
Heat gun	Heat-shrink tubing; a lighter is less controlled
Small screwdriver set	Terminals and enclosure assembly
Drill and step bit	Cable-gland and display-window holes in commercial enclosure
Hot glue gun or silicone sealant	Strain relief and noncritical mechanical support
USB power meter	Optional; helpful during programming/charging
Inline current meter or coulomb meter	Optional; best for measuring real battery runtime
Measuring cup or graduated cylinder	Pump flow calibration in mL per second

Wiring Plan

Battery and pump power

Protected 18650

|
├ 18650 holder → JST-PH lead → Feather BAT connector
└ power switch → fuse → pump positive

Pump negative —————
|
└ MOSFET switched output / drain

MOSFET ground ————— battery negative / Feather ground
ESP32 GPIO ————— MOSFET logic input

Flyback diode

If the MOSFET module does not already contain one:

Pump positive ———|<|—— Pump negative

Striped/cathode end → Pump positive
Unstriped/anode end → Pump negative

Capacitors

470–1,000 μF capacitor:
+ → pump positive after fuse
– → shared battery ground

0.1 μF ceramic capacitor:
directly across pump terminals

Sensors and button

Feather 3.3 V → moisture sensor VCC
Feather GND → moisture sensor GND
Feather ADC → moisture sensor analog output

Float switch:
one terminal → Feather GPIO configured INPUT_PULLUP
other terminal → Feather GND

Setup button:
one terminal → Feather wake-capable GPIO configured INPUT_PULLUP
other terminal → Feather GND

E-paper display

```
Feather 3.3 V → display VCC
Feather GND   → display GND
Feather MOSI  → display DIN
Feather SCK   → display CLK
Spare GPIO    → display CS
Spare GPIO    → display DC
Spare GPIO    → display RST
Spare GPIO    → display BUSY
```

Verify the exact selected Feather pinout and the exact e-paper breakout pin names before soldering.

Software and Development Setup

Install:

```
Visual Studio Code
PlatformIO IDE extension
Wokwi Simulator extension
Git
Python 3
```

Create the project with:

```
Platform: Espressif 32
Board: Adafruit Feather ESP32-S3
Framework: Arduino
```

Suggested starter platformio.ini:

```
[env:adafruit_feather_esp32s3]
platform = espressif32
board = adafruit_feather_esp32s3
framework = arduino

monitor_speed = 115200
board_build.filesystem = littlefs

lib_deps =
    zinggjm/GxEPD2
    adafruit/Adafruit GFX Library
    bblanchon/ArduinoJson
```

Add a web-server library after core watering/safety behavior works:

```
ESP32Async/ESPAsyncWebServer
```

Control Algorithm Requirements

Do not implement automatic watering as one raw sensor threshold. Use a safety-limited state machine.

Required checks before an automatic dose

```
Automatic mode enabled
Sensor reading valid
Reservoir is not low
Battery voltage is above pump-lockout threshold
Pump is not already running
Moisture is at/below dry threshold
Cooldown has expired
Daily pump-time budget remains
```

Safe initial defaults

Sensor wake/sample interval:	45 minutes
Samples per measurement:	20
Initial dry threshold:	30% calibrated relative moisture
Re-arm/wet threshold:	50%
Automatic dose:	3 seconds
Hard maximum one pump run:	8 seconds
Automatic cooldown:	8 hours
Maximum automatic pump time/day:	15 seconds
Soak/recheck delay:	60 minutes
Low-battery warning:	3.50-3.69 V at rest
Pump lockout threshold:	3.45-3.50 V at rest

Calibration

1. Mount the moisture sensor in the real pot at its permanent depth/location.
2. Let the pot reach the dryness point where you would normally water it.
3. Take multiple raw ADC readings and save their average as `dryRaw`.
4. Water normally, let the pot drain and stabilize for roughly 12-24 hours.
5. Take multiple readings and save their average as `wetRaw`.
6. Convert future readings into a relative 0-100% moisture scale.
7. Confirm whether readings increase or decrease when wet; sensor polarity differs.

Mandatory fail-safes

- Pump GPIO must initialize LOW/off immediately at boot.
- Float switch must block pump activation when reservoir is low.
- Every manual or automatic pump command must have a hard time cap.
- A daily runtime cap must block excessive total watering.
- Invalid, out-of-range, or stale sensor readings must block automatic watering.
- Low battery must block pumping before the cell is over-discharged.
- The e-paper/UI must never be required for the safety logic to work.
- Log all watering events, faults, raw sensor readings, and battery state.

Battery Maintenance Plan

Use battery swaps rather than relying on charging the unit at the plant.

```
Cell A installed in Verdant
Cell B charged spare indoors
When battery warning appears:
  swap A → B
  charge A using external 18650 charger
```

Do not use damaged, dented, hot, leaking, swollen, or torn-wrap cells. Do not use salvaged laptop cells. Do not mix cell types in a multi-cell pack.

Suggested e-paper messages:

```
BATTERY: 82% / 3.91 V
BATTERY: LOW — SWAP SOON
BATTERY: CRITICAL — PUMP LOCKED
```

Purchase Checks

Confirm every item below before ordering:

- ☐ Feather accepts a single-cell 3.7 V nominal / 4.2 V maximum Li-ion battery.
- ☐ 18650 holder fits the exact protected/button-top cell selected.
- ☐ JST-PH lead polarity matches the Feather battery socket.
- ☐ Pump genuinely works throughout roughly 3.4–4.2 V, not only at 5 V or 6 V.
- ☐ Pump stall current is below the MOSFET module, fuse, wiring, and cell capabilities.
- ☐ MOSFET driver works with a 3.3 V ESP32 GPIO signal.
- ☐ The flyback diode is included or separately purchased.
- ☐ E-paper display works at 3.3 V logic and has a compatible Arduino library.

- [] Enclosure is IP65 minimum and glands match the cable outer diameters.
- [] Enclosure dimensions allow for the Feather, display, battery holder, cable bends, standoffs, and fuse.
- [] No battery/pump voltage will ever connect directly to an ESP32 GPIO.

Do Not Buy for v1

- Relay module
- Standard Arduino Uno/Nano
- Generic ESP32 DevKit as the final battery controller
- 12 V pump or 12 V power supply
- 5 V-only pump unless you intentionally add a boost converter
- Resistive two-prong soil moisture probe
- Tri-color e-paper display
- Always-on OLED/LCD
- Loose unprotected/recycled 18650 cells
- Improbable-capacity single 18650 cells, such as 9,900 mAh or 15,000 mAh claims
- A basic TP4056 charger as the only charge/power-path solution while operating the device during charging
- A bare 3D-printed enclosure as the only water barrier

Build and Test Order

1. Develop C++ controller logic in PlatformIO and simulate the high-level control behavior in Wokwi.
2. Flash the Feather by USB and prove deep-sleep/wake behavior before connecting pump hardware.
3. Wire the MOSFET driver and test with an LED or multimeter before the real pump.
4. Connect pump to a measuring cup; record current, startup behavior, and mL/sec flow rate.
5. Add float switch; confirm it blocks the pump immediately when reservoir is low.
6. Add soil sensor; collect dry/wet readings and calibrate it in the actual potting mix.
7. Build the e-paper display and verify it survives controller deep sleep with the last status visible.
8. Assemble the enclosure with all glands/window installed but no electronics inside; put tissue paper inside and perform a gentle spray test.
9. After the empty enclosure passes the spray test, install electronics and test in a catch tray.
10. Run in manual/supervised mode for at least one week.

11. Enable conservative automatic watering only after confirming dose size, sensor calibration, float switch behavior, and battery limits.

Final Checklist

- ☐ Pump defaults off after boot/reset
- ☐ Manual pump run has hard maximum duration
- ☐ Reservoir-low switch blocks pump
- ☐ Daily runtime cap works
- ☐ Battery lockout works
- ☐ Sensor-invalid lockout works
- ☐ Flyback protection is installed
- ☐ Pump wiring is fused
- ☐ Battery polarity was checked with multimeter
- ☐ Pump flow rate was measured
- ☐ Display is hibernated after update
- ☐ Electronics enclosure passed tissue-paper spray test
- ☐ Tubing never passes through the dry electronics enclosure
- ☐ Enclosure is mounted above reservoir and uses downward-facing glands
- ☐ Device has run supervised in a catch tray before unattended mode